How pairing digital twin technology with AI could boost buildings' emissions reductions

- **Around the world, new regulations are targeting building emissions. From this year, for example, New York City will impose strict caps on more than 40,000 buildings to combat greenhouse gas emissions.**
- **Digital twins are virtual building replicas, which enable real-time analysis of climate-related data, predictive maintenance capabilities and cost savings.**
- **In the future, digital twins could lead to cognitive twins – intelligent systems capable of predictive analysis, anomaly detection and autonomous operation. This could enhance building efficiency and occupant comfort.**

This year, New York City will start enforcing the most ambitious building performance standard legislation in the world. Local Law 97 (LL97), enacted in 2019, caps emissions from buildings over 25,000 square feet and applies to more than 40,000 buildings.

The number of jurisdictions adopting building performance standards in the United States has nearly doubled since 2020. Buildings account for more than 30% of all greenhouse gas emissions in the US. In New York City, a densely populated area, buildings contribute nearly 70% of overall emissions.

As the number of building performance standards grows to combat these emissions, simplifying and automating environmental, social and governance (ESG) reporting, and enabling implementation of energy conservation measures, is becoming more important.

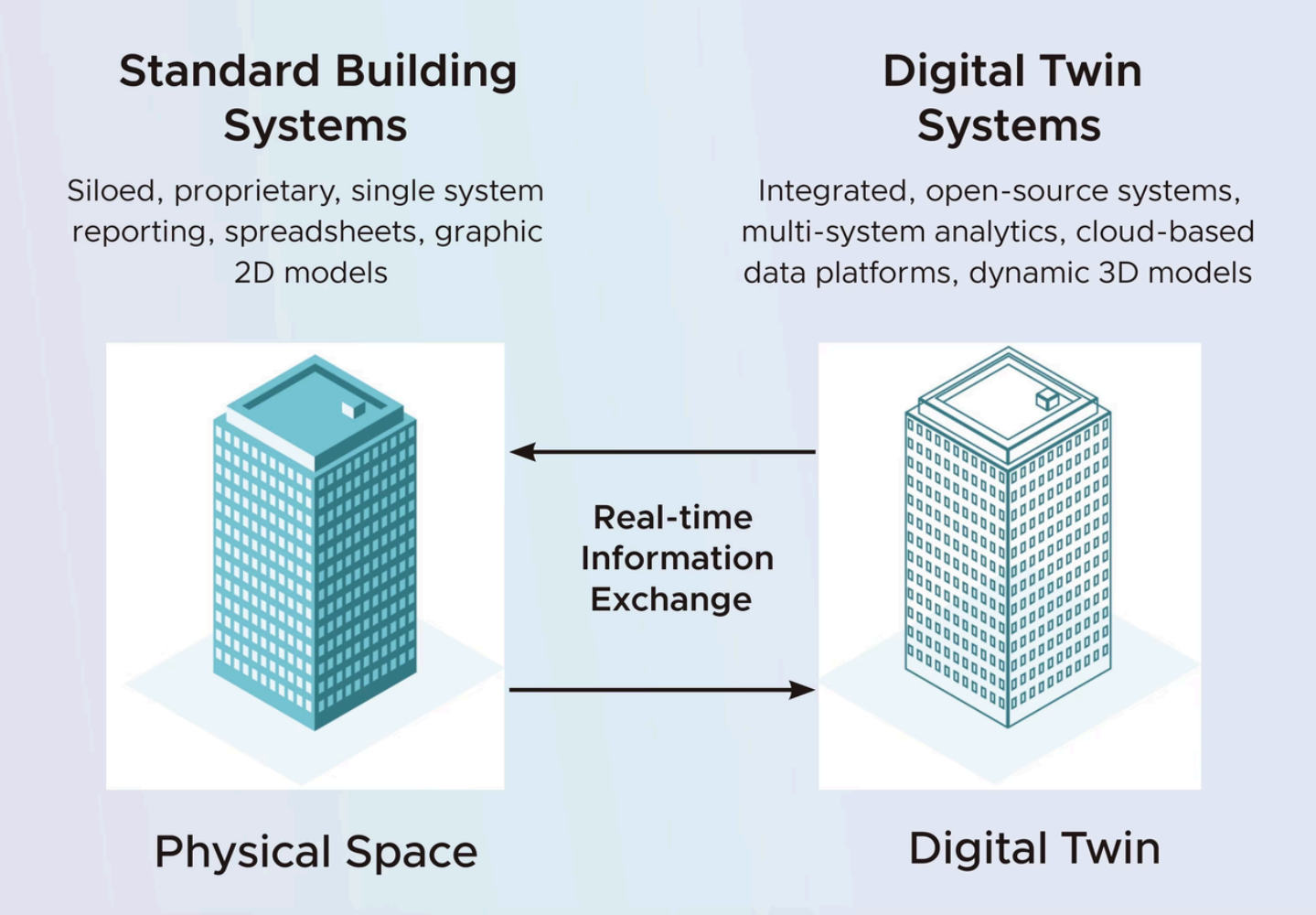
But existing building management systems and ESG reporting tools are limited in scope and operate in siloes. Building management systems are also often proprietary, so they are incompatible with technologies and equipment from different vendors. As a result, ESG reporting is highly resource intensive. Collecting Scope 3 data – emissions information from suppliers and vendors – is also incredibly challenging.

Digital twin technology paired with artificial intelligence (AI) could be a promising solution. This could help building management systems realise significant cost savings and operating efficiency using predictive maintenance, scenario testing, risk management and automated real-time reporting.

What is a digital twin?

A digital twin is a virtual replica of a physical object such as a building, including all of its equipment and operating systems. Smart meters and equipment sensors transmit data wirelessly to the digital twin via the cloud, so accurate, fine-grained data can be captured and tracked in real time on a single platform.

Such platforms can collect and analyse vast amounts of data from multiple sources across a campus, portfolio of buildings or supply chain. This means energy, emission, water, waste, occupancy, air quality and transportation data can be tracked, traced and monitored. This high-fidelity, highly synchronized data is then fed into Al systems to support informed and precise decision making. An organization's Scope 1, 2 and 3 emissions reports can then be generated in real-time to support compliance with benchmarking or reporting frameworks.



Global home furnishings company IKEA is already using this technology. The big box furniture store has digitalized 37 of its retail stores in East Asia in only nine months to improve energy efficiency and reduce greenhouse gas emissions. It was able to connect 7,000 different data points from various building and energy management, and Internet of Things (IoT) systems to a single digital twin.

Ikea's digital twin represented nearly 42 million square feet of space and 6,000 pieces of heating ventilation and air conditioning (HVAC) equipment from 10 different manufacturers. Energy consumption by the HVAC central supply system was reduced by 30% with annual energy cost savings estimated to be in the millions.

The next step: cognitive digital twins

Over time, buildings with AI-driven digital twin systems will evolve to become "cognitive" by learning from existing processes and patterns. They will detect anomalies, recommend corrective action and predict faults before they occur.

These “intelligent” systems would provide a critical support service to facilities staff and extend equipment life, improving return on investment and operating costs. By monitoring and maintaining room-level heating, cooling and air quality, these systems will also vastly improve building occupants’ health and reduce sick days.

As the level of automation advances, operating efficiency is fine-tuned. This reduces energy use, labour, emissions and operating costs. In the future, cognitive buildings might resemble Tesla’s self-driving cars – essentially running themselves with human oversight and the ability to override.

Cognitive buildings designed to test scenarios in the virtual world using digital twin technology can minimize risk and prevent costly mistakes in the real world. This might include anything from estimating how much battery storage should accompany the installation of solar panels, to modelling how the panels will perform during an extreme weather event, or gauging how electrification will affect your electrical load and energy bills. Complex decision making becomes more accurate, data-driven and collaborative as a result.
For example, retrofits of critical systems such as HVACs are traditionally carried out with little to no visibility prior to the projects, and often without any visibility after the fact. Digital twin technology gives building owners the opportunity to “test before they invest” in retrofits. Not only does it allow them to build a baseline for the cost, energy performance and carbon emissions impacts, it enables them to simulate how those numbers might look post-retrofit – before they even spend a dollar.